

Lab: Social Neuroscience and Mirror Systems

Learning Goal: Analyze social brain circuits through Active Inference, evaluating mirror neuron accounts and language processing.

Part 1: Social Brain Network Mapping

Exercise: For a specific social scenario (someone telling you an emotional story), map the activation of each social brain region:

Brain Region	What It Computes	Active During Story Because...	Prediction Generated
mPFC (dorsal)	Deliberate mentalizing	Inferring the storyteller's mental state	"She feels sad about the loss"
mPFC (ventral)	Automatic social prediction	Rapidly predicting emotional tone	"The next part will be even sadder"
TPJ (right)	Self-other distinction	Distinguishing your reaction from hers	"I feel empathy, but she feels grief"
STS	Action/gesture interpretation	Reading body language and facial expression	"Her downcast eyes signal sorrow"
Anterior insula	Empathic resonance	Mapping her interoceptive state onto yours	"I feel a tightness in my chest"
FFA	Face processing	Reading minute facial expressions	"Micro-expression of pain around eyes"
Temporal pole	Narrative/semantic context	Tracking the story's meaning	"This connects to her broader life story"

Write a 200-word analysis: How do all these regions coordinate to produce the experience of "understanding someone"?

Write your response here

Part 2: Mirror Neuron Debate

Learning Goal: Evaluate competing accounts of mirror neuron function.

Exercise: Compare the original and Active Inference accounts:

Feature	Original Mirror Theory	Active Inference Account
Mechanism	Direct motor matching — simulation	Predictive model generating same predictions for self and observed actions
Required hardware	Dedicated mirror neuron system	Any sufficiently sophisticated generative model
Experience effects	Not well explained	Naturally explained (better model → stronger predictions)
Context modulation	Not well explained	Naturally explained (top-down context shapes predictions)
Lesion effects	Should abolish action understanding	Should degrade but not abolish (other routes to understanding)
Developmental trajectory	Innate or early-developing	Develops with experience and model refinement

Write a 300-word analysis: Which account is better supported by current evidence? What experiment could definitively distinguish them?

Write your response here

Part 3: Language Prediction Experiment

Learning Goal: Analyze language processing as hierarchical prediction.

Exercise: Consider the following sentence: "The cat sat on the ____."

1. At the phonological level, what consonant cluster does the brain predict will start the next word?
2. At the lexical level, what specific words are highly predicted? (mat, rug, chair, floor...)
3. At the syntactic level, what word class is predicted? (noun)
4. At the discourse level, what semantic field is predicted? (furniture/surfaces)
5. If the sentence ends with "moon," what prediction errors fire at each level?
6. Which ERP components (N400, P600) would you expect? Why?

Now design a more complex example using pragmatic prediction: "It was my birthday, and everyone had gathered. I opened the box, and inside was a ____."

Write your response here

Part 4: Interbrain Synchronization Analysis

Learning Goal: Analyze neural coupling between interacting minds.

Exercise: Design a hyperscanning experiment:

1. **Scenario:** Choose an interactive scenario (conversation, cooperative game, teaching)
2. **Measurement:** What brain signals would you record from both participants? (EEG? fNIRS? fMRI?)
3. **Synchronization metric:** How would you quantify brain-to-brain coupling?
4. **Active Inference predictions:** What conditions would increase synchronization (shared priors, aligned models) vs. decrease it (disagreement, different priors)?
5. **Control condition:** How would you distinguish genuine social coupling from shared stimulus-driven activity?

Write your response here

Part 5: Reflection

In 300 words, reflect: Active Inference treats social cognition as inference about other generative models — "thinking about thinking." But is this what social interaction *feels* like? Phenomenologically, understanding another person feels immediate and embodied, not computational. Can Active Inference account for this phenomenological directness, or does it miss something essential about social connection?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Network mapping	Social brain circuit coordination
2	Theory comparison	Mirror neuron debate
3	Hierarchical analysis	Language prediction levels
4	Experiment design	Interbrain synchronization
5	Phenomenological reflection	Social connection vs. inference